Relational Database Schema (ERD), Causal Time Indexes & 12-D Graph Feature View

TABLE: entities_master

PK	entity_id	BIGINT / UUID
	entity_type	VARCHAR(32)
	kyc_risk_tier	SMALLINT (1-5)
FK	country_iso	CHAR(2) -> Jurisdictions
	created_timestamp	TIMESTAMP WITH TZ
	is_pep_flag	BOOLEAN DEFAULT FALSE
	initial_balance	NUMERIC(18, 4)

1 : N (Source)

1 : N (Target)

TABLE: streaming_transactions

PK	transaction_id	BIGINT / UUID
FK	source_entity_id	BIGINT -> entities
FK	target_entity_id	BIGINT -> entities
	amount_usd	NUMERIC(18, 4)
	timestamp_utc	TIMESTAMP WITH TZ
	payment_rail	VARCHAR(16) [SWIFT/bKash]
	channel_code	VARCHAR(16) [ATM/API/WIRE]
	conformal_label	SMALLINT [0=Benign, 1=Illicit]

1 : [illegible]

TABLE: high_risk_jurisdictions

PK	country_iso	CHAR(2)
	country_name	VARCHAR(64)
	fatf_status	VARCHAR(24) [Grey/Black]
	basel_aml_index	NUMERIC(4, 2)
	sanction_authority	VARCHAR(32) [OFAC/UN]
	effective_date	DATE

DUCKDB / POSTGRESQL TIME-CAUSAL COMPOSITE INDEXES & MATERIALIZED 12-D INVARIANT VIEW

Composite Causal Indexes (Zero Lookahead):

CREATE INDEX idx_source_time
ON streaming_transactions(source_entity_id, timestamp_utc DESC)
CREATE INDEX idx_target_time
ON streaming_transactions(target_entity_id, timestamp_utc DESC)
Execution Guarantee:
O(log |E|) 2-hop causal ego-graph extraction in <0.18 ms
Lookahead Invariant:
Strictly t_e <= t_eval (eliminates future edge leakage)

CREATE VIEW v_canonical_invariants_12d AS (Mapping to Directed Multigraph):

1. **Flow Regularity:** $\Phi_{\text{flow}} = \ln(1 + \sum A_{\text{out}}/(\sum A_{\text{in}} + \epsilon))$
2. **Hawkes Arrival Rate:** $\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha \cdot e^{-\beta(t - t_i)}$
3. **Directed Causal Taints:** $s_{\text{fwd}} \in [0, 1]$ **(Dispersion),** $s_{\text{bwd}} \in [0, 1]$ **(Aggregation)**
4. **Graph Multi-Moments:** $[\mu_A, \sigma_A^2, \gamma_A, \kappa_A] \parallel [\deg_{\text{in}}, \deg_{\text{out}}, \nu_A, \nu_{io}] \rightarrow \mathbb{R}^{12}$